How to develop a chemical engineering process

Chemical engineer role

- Chemical engineers translate processes developed in the lab into practical applications for the commercial production of products and then work to maintain and improve those processes.
- They rely on the main foundations of engineering: Math, Physics, And Chemistry (though Biology is playing an increasing role).

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- The main role of chemical engineers is to design and troubleshoot processes for the production of chemicals, Petrochemicals, foods, pharmaceuticals, and biologicals, just to name a few.
- They are most often employed by large-scale manufacturing plants to maximize productivity and product quality while minimizing costs.

The aerospace, automotive, biomedical, electronic, environmental, medical, and military industries seek the skills of chemical engineers in order to help develop and improve their technical products, such as:

- ❖Ultrastrong fibers, fabrics, and adhesives for vehicles
- ❖Biocompatible materials
- ❖Films for optoelectronic devices

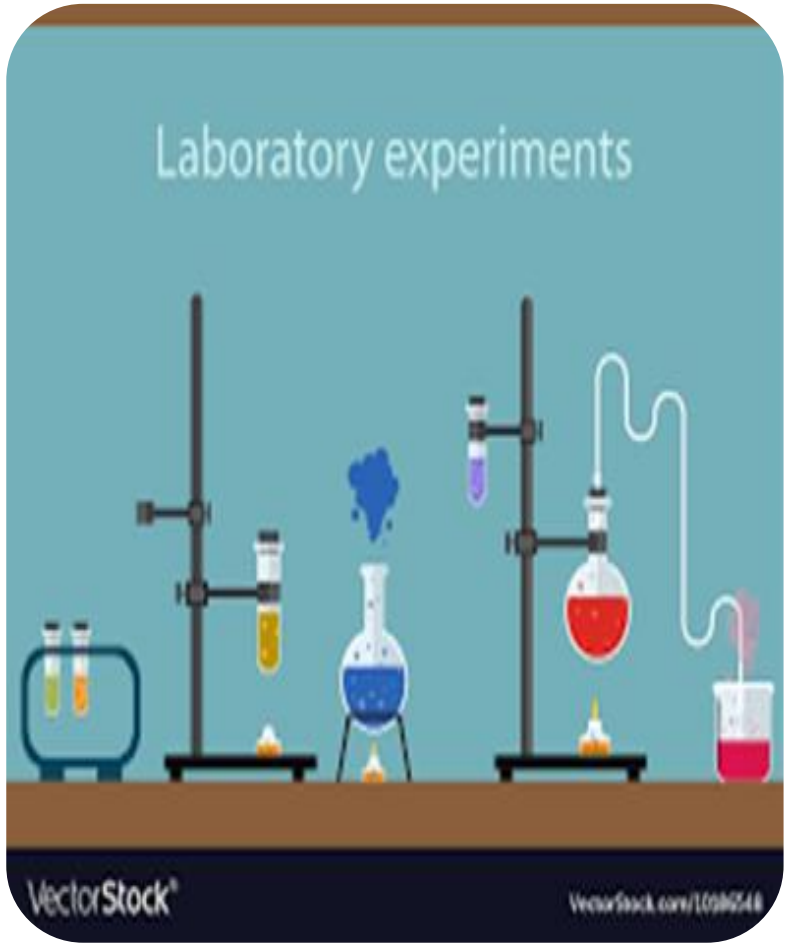
Chemical engineers work in almost every industry and affect the production of almost every article manufactured on an industrial scale.

Some typical tasks include:

- Ensuring compliance with health, safety, and environmental regulations
- Conducting research into improved manufacturing processes
- Designing and planning equipment layout
- Incorporating safety procedures for working with dangerous chemicals
- Monitoring and optimizing the performance of production processes
- Estimating production costs

Chemical process design and development

- **The process from lab to industry**





Production of Ammonia plant

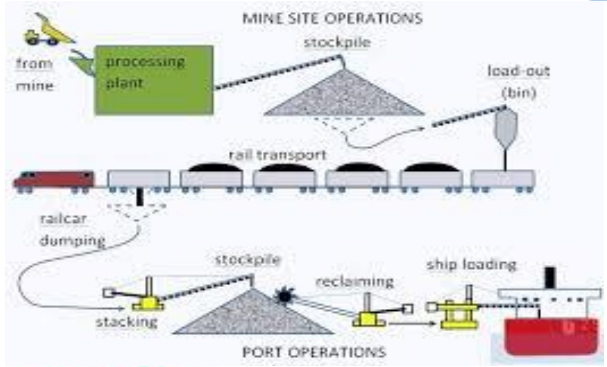
- Assume the following reaction

$$N_2 + 3H_2 \rightarrow 2NH_3$$

- ***STEPS for process development***
- ***1- Raw materials type (S,L or G)***
- *N.B.*
- *(For above reaction all raw materials are gases)*
- *For Raw Materials First Step Is Storage And Handling*

Storage and Handling of Materials

storage







Solids are stored in piles either indoor or outdoor according to its type

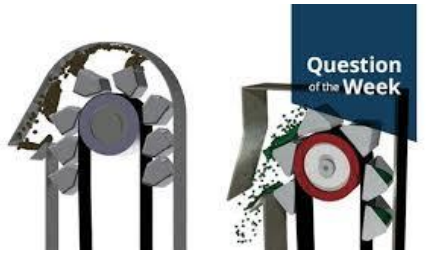
Also store in silos or in big bags

Liquids usually store in cylinders either under pressure of atmospheric conditions.

Gases usually store under pressure in either cylindrical or spherical vessels

handling

Using belt or bucket elevators or dumped directly to the reactor



Handling using pumps to drive the liquid

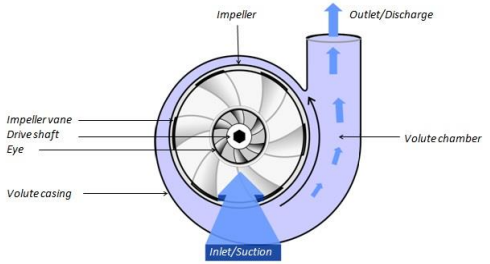


Figure 2. Volute case design

Handling using compressors to drive the gas



https://youtu.be/l3C2r4I5o8Y?t=121

- https://youtu.be/XrIVG_SqR_U?t=135

Problem 1:

- It is required to store 5000 kg natural gas in a spherical tank under STP conditions (T=273K and P= 1bar).

1- Calculate the storage area required

Solution

- As $n = \frac{W}{M}$
- Thus: n = $\frac{5000}{16}$ = 312.5 kmole = 312500 mole
- As 1 mole of any gas occupies 22.4 liter at STP
- The total volume required = 312500 * 22.4 = $7*10^6$ liter = $7*10^3$ m^3
- As the volume of a sphere = $4 \pi R^3 /3$
- R = $(3*7000/ 4 \pi)^{1/3}$ = 11.86 m
- $A=4\pi r^2$
- A = $4 * \pi * (11.86)^2$ = 1766.68 m^2

1. What type of reactor should be used? A long pipe? A large tank? Several smaller tanks? An extremely large test tube? How large? Made of what? Does it have to be heated? If so, how much and how? With an electrical heater inside or outside the reactor? By passing a hot fluid through a heating coil in the reactor? By heating the reactants before they get into the reactor? Does the reaction supply its own heat, so that heating is needed only for startup? If so, can the reactor "run away" and possibly explode? Should control measures be introduced to prevent this? What kind?







Chemical Process Reactors



- A chemical reactor is an enclosed volume in which a chemical reaction takes place.
- In chemical engineering, it is generally understood to be a process vessel used to carry out a chemical reaction



- The most common **basic types** of chemical **reactors are**

➢**Tanks** (where the reactants mix in the whole volume) and

➢**Pipes or tubes** (for laminar flow reactors and plug flow reactors)

- **Both types can be used as**

➢**Continuous reactors or**

➢**Batch reactors,**

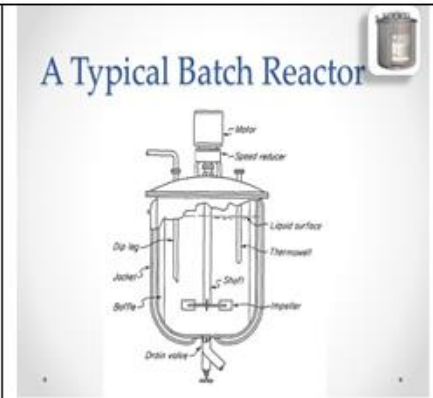
and either may accommodate one or more solids (reagents, catalysts, or inert materials), but the reagents and products are typically fluids (liquids or gases).

- **Batch or Continuous Processes**

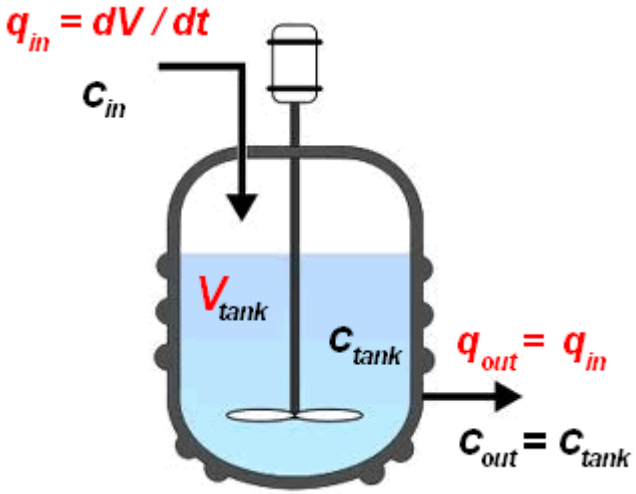
- **A Batch Process**
  - is one in which the reactants are placed into a vessel and allowed to react with each other.
  - When the reaction is complete, the product is removed for purification, and the process is repeated for a fresh batch of materials.

- **A continuous process**
  - Is one in which the raw materials are constantly fed into the plant, reacting as they move through the equipment to give a continuous flow of product.

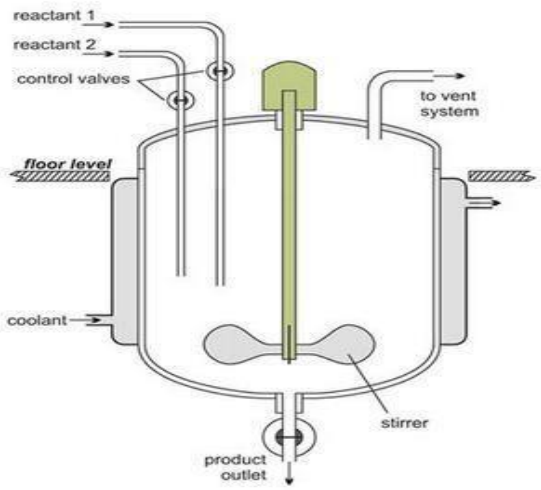
- **A semi-continuous process**
  - is a combination of the first two.

	Batch	Continuous
Examples	Production of dyes and steels. Pharmaceuticals such as paracetamol.	Iron Sulphuric Acid
Advantages	Cost effective for small quantities. Range of products can be made in one vessel.	More easily automated. Lower numbers of process operators needed.
Disadvantages	Increased risk of cross-contamination. Larger work-force required.	High capital costs to build the plant. Not cost effective when running below capacity.
Shape		

- Reactors in continuous processes are typically run at steady-state, whereas reactors in batch processes are necessarily operated in a transient state.
- When a reactor is brought into operation, either for the first time or after a shutdown, it is in a transient state, and key process variables change with time.



Continuous Reactor



Batch Reactor

Types of Reactors

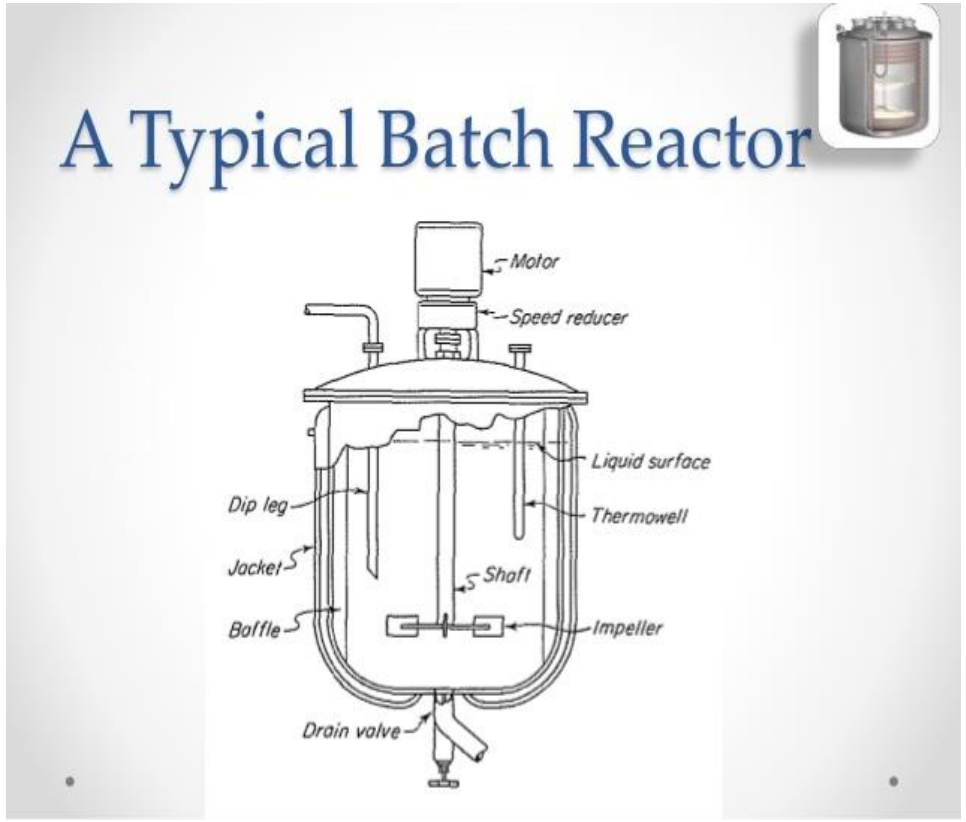


- There are three idealized models used to estimate the most important process variables of different chemical reactors:
- Batch reactor model
- The simplest type of reactor is a batch reactor. Materials are loaded into a batch reactor, and the reaction proceeds with time.
- A batch reactor does not reach a steady state, and control of temperature, pressure and volume is often necessary.



Batch Reactor (Continues)

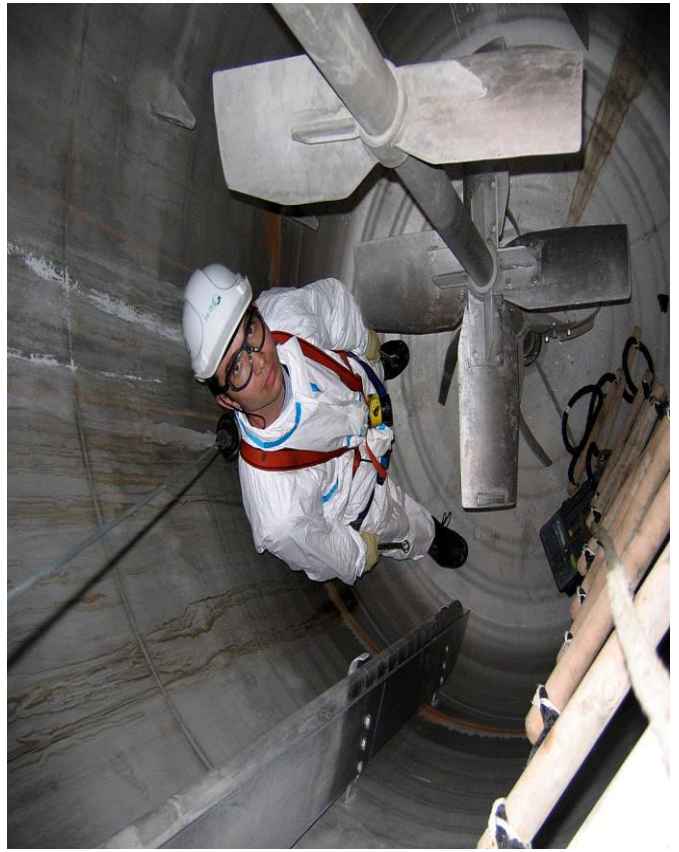
- Many batch reactors therefore have ports for sensors and material input and output.
- Batch reactors are typically used in small-scale production and reactions with biological materials, such as in brewing, pulping, and production of enzymes. One example of a batch reactor is a pressure reactor.



- https://youtu.be/ftnLJ6VDwS8?t=11

Continuous stirred-tank reactor model (CSTR)

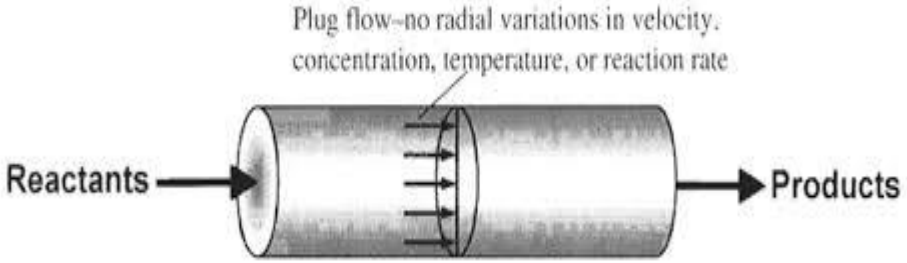
- In a CSTR, one or more fluid reagents are introduced into a tank reactor which is typically stirred with an impeller to ensure proper mixing of the reagents while the reactor effluent is removed.
- Dividing the volume of the tank by the average volumetric flow rate through the tank gives the space time, or the time required to process one reactor volume of fluid. Using chemical kinetics, the reaction's expected percent completion can be calculated.



Some important aspects of the CSTR:

- At steady-state, the mass flow rate in must equal the mass flow rate out, otherwise the tank will overflow or go empty (transient state).
- While the reactor is in a transient state the model equation must be derived from the differential mass and energy balances.
- The reaction proceeds at the reaction rate associated with the final (output) concentration, since the concentration is assumed to be homogenous throughout the reactor.
- Often, it is economically beneficial to operate several CSTRs in series. This allows, for example, the first CSTR to operate at a higher reagent concentration and therefore a higher reaction rate. In these cases, the sizes of the reactors may be varied in order to minimize the total capital investment required to implement the process.
- It can be demonstrated that an infinite number of infinitely small CSTRs operating in series would be equivalent to a PFR.

Plug flow reactor (PFR)



- In a PFR, sometimes called continuous tubular reactor (CTR).
- one or more fluid reagents are pumped through a pipe or tube.
- The chemical reaction proceeds as the reagents travel through the PFR. In this type of reactor, the changing reaction rate creates a gradient with respect to distance traversed; at the inlet to the PFR the rate is very high, but as the concentrations of the reagents decrease and the concentration of the product(s) increases the reaction rate slows.
- **Some important aspects of the PFR:**

➢Reagents may be introduced into the PFR at locations in the reactor other than the inlet. In this way, a higher efficiency may be obtained, or the size and cost of the PFR may be reduced.

➢A PFR has a higher theoretical efficiency than a CSTR of the same volume. That is, given the same space-time (or residence time), a reaction will proceed to a higher percentage completion in a PFR than in a CSTR.

➢This is not always true for reversible reactions.

- https://youtu.be/yPOVF2IzBJg?t=13

Key process variables include:

- Residence time (t)
- Volume (V)
- Temperature (T)
- Pressure (P)
- Concentrations of chemical species (C_1, C_2, C_3, ... C_n)
- Heat transfer coefficients (h, U)

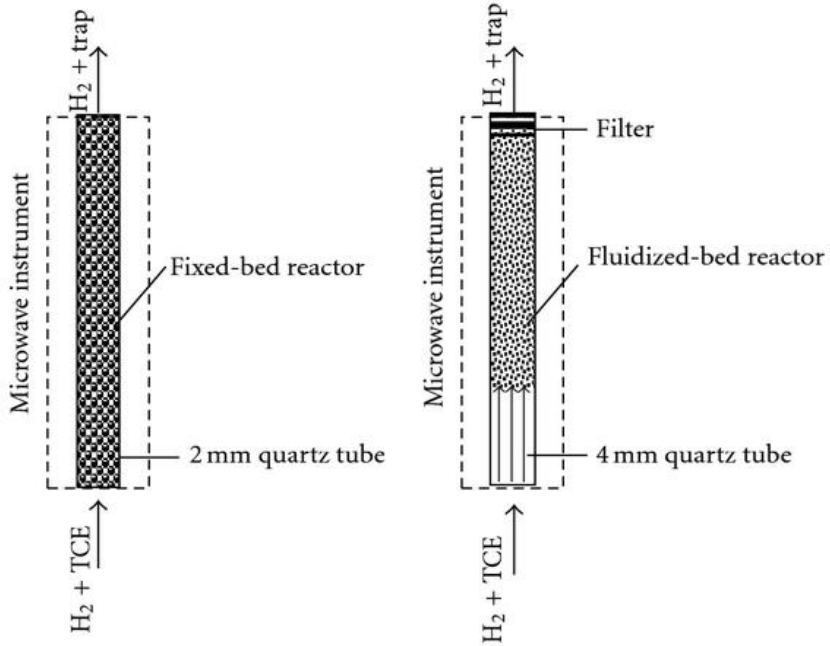
Types of Tubular reactors

- A tubular reactor can often be:
- Packed bed. In this case, the tube or channel contains particles or pellets, usually a solid catalyst. The reactants, in liquid or gas phase, are pumped through the catalyst bed. A chemical reactor may also be a

- Fluidized bed; see Fluidized bed reactor.

Particles are fluidized to achieve

Higher surface area



Heat of reactions



Coil



- **Chemical reactions occurring in a reactor may be:**
- Exothermic, meaning giving off heat, or
- Endothermic, meaning absorbing heat.
- **SO**
- A tank reactor may have a **cooling or heating jacket** or **cooling or heating coils** (tubes) wrapped around the outside of its vessel wall to cool down or heat up the contents, while tubular reactors can be designed like heat exchangers if the reaction is strongly exothermic, or like furnaces if the reaction is strongly endothermic.



Jacket

- Calculate the area of a reactor given the following information:

➢Flow rate of reactant = 200 kmole/min

➢Average molecular mass of reactants = 32

➢Average density = 1.82 g/cm^3

➢The required contact time = 25 min.

➢The reactor is cylindrical shape with height of 3m

- **Solution**
- V = Q*t , Where Q is the volumetric rate
- Q = mass/density = moles*molecular mass/density
- Q = 200000*32/1.82= 3516483 cm^3/min = 3.516 m^3/min
- V = 3.516 * 25 = 87.91 m^3
- V = A * H
- A = V/H =87.91/3 = 29.3 m^2

V= m3/s

n ,m , v

n = m/M

m = density * V

n = density*V/M

Q= V = n*M/density

Vol of reactor = Q*t

- **in case of solution mixture the**
- **Average molecular weight:**
- $M_{av} = \Sigma\, y_i\, M_i$

y_i = mole fraction = moles of i/total moles

M_i = molecular weight of component i

- **Average density for mixtures**
- $\rho_{av} = \Sigma\, y_i\, \rho_i$

y_i = mole fraction = moles of i/total moles

ρ_i = density of component i